Towards Trustworthy AI in Official Statistics: Methodological Challenges and Research Perspectives from Istat

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Abstract. Artificial Intelligence (AI) is increasingly transforming the production and dissemination of Official Statistics. While AI technologies offer significant opportunities to improve efficiency, exploit new data sources, and enhance access to statistical information, their adoption raises critical methodological challenges related to validation, transparency, and privacy protection. This paper examines the role of trustworthy AI in Official Statistics, focusing on the methodological implications of integrating AI-based methods into statistical production processes. After outlining AI opportunities across the phases of the statistical production process, the paper identifies key challenges related to validating AI-based methods, adapting quality frameworks, and the responsible use of data. Building on ongoing research at the Italian National Institute of Statistics (Istat), the paper proposes an integrated approach that combines methodological research, institutional governance, capacity building, and collaboration with public research infrastructures. This approach provides a practical framework for supporting the responsible and sustainable adoption of AI in Official Statistics while preserving statistical quality and public trust.

Keywords: Artificial Intelligence, Trustworthy AI, Official Statistics

1 Introduction

Over the past decade, the environment in which National Statistical Institutes (NSIs) operate has undergone a profound transformation. The digitalization of society and the rapid progress of Artificial Intelligence (AI) are reshaping how data is created, accessed, and used. Data ecosystems have become increasingly complex, and expectations for Official Statistics have grown [1]. Users now demand more timely releases, greater levels of detail, and simpler ways to access and interact with statistical information.

However, the adoption of AI in Official Statistics also raises important methodological and institutional challenges. NSIs operate within established quality frameworks and legal constraints designed to ensure reliability, transparency, and the protection of privacy and confidentiality, as reflected in the European Statistics Code of Practice [2].

AI-based approaches, particularly when relying on complex or black-box models, may introduce new risks related to validation, interpretability, data protection, and potential biases in data or algorithms. Addressing these risks while ensuring alignment with the core principles of Official Statistics—such as methodological soundness, transparency, and public trust—represents a central challenge for statistical organizations [3].

This paper examines the opportunities and methodological challenges of adopting trustworthy AI in Official Statistics, drawing on ongoing research at Istat. Specifically, it focuses on key issues such as the validation of AI-based methods, the adaptation of quality frameworks to AI-driven processes, and the responsible use of data in statistical production. It also presents examples of current research at Istat aimed at integrating AI techniques into statistical processes while preserving statistical quality and reliability.

2 AI Opportunities in Official Statistics

Artificial Intelligence is increasingly being utilized by National Statistical Institutes to support various activities throughout the statistical production process. As shown in Figure 1, these applications span data collection, statistical processing, and data dissemination.

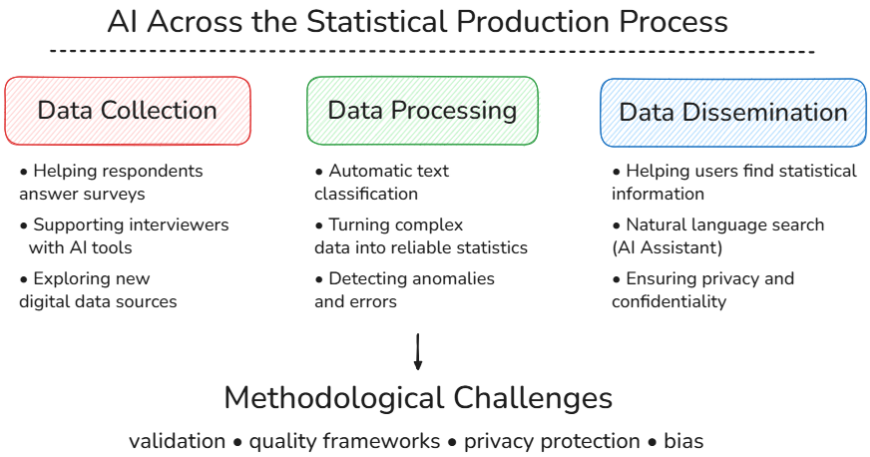

Fig. 1. Examples of how Artificial Intelligence can support different phases of the statistical production process and the related methodological challenges.

Data collection. AI technologies can enhance data collection in multiple ways. Intelligent systems can assist respondents in completing questionnaires, support interviewers through smart tools, and facilitate the integration of administrative and non-traditional data sources into production processes. These approaches

can reduce response burden and data collection costs, improve data quality, and broaden the range of information available for statistical processing.

Data processing. AI techniques enable the transformation of complex and heterogeneous data into reliable statistical information. Applications include automatic text classification, detection of non-sampling errors, and imputation of missing values in large and complex datasets. These methods allow statistical offices to process large volumes of data more efficiently while maintaining established statistical quality standards.

Data dissemination. AI provides new opportunities to improve access to Official Statistics. Natural language interfaces and AI assistants can help users more easily retrieve relevant statistical information, generate personalized tables and visualizations, and interact with statistical content more intuitively. However, these advancements must still adhere to the principles of privacy and confidentiality that are fundamental to Official Statistics.

Overall, the adoption of AI across these phases reflects a broader shift towards more flexible, scalable, and data-driven statistical production systems that can leverage both traditional and non-traditional data sources.

3 Methodological Challenges for Trustworthy AI

While AI offers significant opportunities across the statistical production process, its adoption in Official Statistics also introduces important methodological challenges. Recent work within the international statistical community has highlighted several critical issues related to the validation of AI-based methods, the adaptation of quality frameworks, and the responsible use of data in AI-driven statistical systems [1, 4, 5].

Validation and statistical reliability. A key challenge is validating AI-based methods used in statistical production. Many AI models rely on complex learning mechanisms that can be opaque, with outputs that may vary depending on the training data, model parameters, or prompting strategies. To ensure robustness, reproducibility, and stability of results, it is essential to develop validation procedures that adhere to established statistical quality standards.

Quality frameworks for AI-driven statistics. Existing quality frameworks for Official Statistics were largely developed for structured data sources and traditional production pipelines. The increasing use of AI methods and non-traditional data sources requires adapting these frameworks to account for new sources of uncertainty and error. These include the probabilistic nature of AI models, dependence on training data, and potential biases in both data and algorithms. In this context, quality assessment must evolve to incorporate new dimensions, such as evaluating model performance under different data conditions, assessing the robustness of results to changes in training data or model configurations, and detecting and mitigating systematic biases. This also requires defining appropriate quality indicators, monitoring mechanisms, and methodological guidelines tailored to AI-assisted statistical production.

Privacy and responsible data use. The use of AI in Official Statistics raises critical issues related not only to privacy protection but also to the operational context in which AI systems are deployed. In particular, the increasing reliance on externally developed models, cloud-based infrastructure, and API-based services introduces new risks regarding data confidentiality, data control, and compliance with legal and institutional constraints. The use of sensitive data within AI systems requires careful consideration of where and how data is processed.

While AI models deployed in controlled, on-premises environments may be compatible with existing confidentiality requirements, using external APIs or shared cloud services may raise significant concerns, depending on data residency, access policies, and system architecture. Addressing these challenges requires robust data governance frameworks, clear rules governing the allowed use of AI technologies, and technical safeguards to ensure that sensitive data remains protected throughout the processing pipeline. These aspects are essential to ensure that AI-supported processes remain aligned with the fundamental principles of Official Statistics, particularly confidentiality, security, and institutional accountability

4 An Institutional Ecosystem for Trustworthy AI at Istat

Building on the challenges outlined above, Istat is addressing AI adoption through a combination of applied research, experimental projects, and strategic collaborations with other research bodies. These activities aim to explore how AI techniques can be integrated into statistical production while preserving statistical quality, transparency, and reliability, as discussed in recent methodological contributions developed within the Official Statistics community [6]. Initiatives are largely coordinated within the Directorate for Methodology and focus on testing AI methods in real statistical contexts, assessing their methodological implications, and identifying the conditions under which they can be safely integrated into production processes.

A primary area of work concerns AI-supported statistical production. Ongoing projects investigate the use of Machine Learning (ML) techniques for automatic classification tasks, such as coding economic activities, industrial products, and causes of death. Recent work on product classification for PRODCOM surveys highlights the potential of AI-driven approaches for transforming web-based or textual data into structured statistical variables [7].

A second area of research investigates the use of new and non-traditional data sources for statistical purposes. Current experiments include analyzing enterprise websites, Earth observation data, Mobile Network Operators data, and patent documents, with the aim of broadening the informational basis of Official Statistics and enabling new forms of statistical measurement [8, 9]. These data sources pose significant methodological challenges, such as issues with data quality, coverage, and interpretability. AI techniques can assist in extracting structured information from these complex sources, but their application re-

quires careful methodological evaluation to ensure that the resulting indicators meet official statistics standards [4].

Research is also being conducted on synthetic data generation, exploring how AI-based techniques can complement traditional statistical disclosure control methods to reduce disclosure risk while preserving analytical utility. Beyond research activities, Istat is establishing institutional mechanisms to promote responsible use of AI. These include internal guidelines and methodological recommendations to ensure that AI-based applications adhere to the principles of Official Statistics. Governance plays a key role in coordinating experimental activities, supporting methodological validation, and ensuring that innovation is implemented in a controlled and sustainable way.

A further component concerns collaboration with public research infrastructures. At the national level, partnerships with organizations such as CINECA provide access to high-performance computing environments for large-scale experimentation with advanced AI models. At the international level, Istat collaborates with other NSIs within initiatives funded by Eurostat and the European Statistical System, focusing on topics such as AI/ML for Official Statistics [10], the integration of mobile phone data with traditional surveys [11], and the use of online job advertisements for labour market analysis [12].

Finally, the adoption of AI in Official Statistics requires strengthening internal competencies. Istat is therefore investing in capacity-building initiatives in areas such as machine learning, data science, and AI governance. Training and knowledge-sharing activities support the diffusion of these skills across the Institute, ensuring that AI systems can be effectively understood, evaluated, and used by statistical experts.

5 Conclusions

Artificial Intelligence offers significant opportunities to improve the use of non-traditional data sources and to enhance the production and dissemination of Official Statistics. However, its adoption must remain aligned with the fundamental principles of statistical systems, especially methodological soundness, transparency, and confidentiality protection.

The main challenge for National Statistical Institutes is not just adopting AI technologies but ensuring that their integration into statistical processes remains trustworthy and methodologically sound. Istat's experience shows how this goal can be achieved through an integrated approach that combines methodological research, experimental activities, institutional governance, and collaboration with public research infrastructures. Moving forward, more work is needed to develop validation protocols for AI-based methods, adapt statistical quality frameworks to AI-driven processes, and create approaches to evaluate robustness, bias, and uncertainty. Ongoing collaboration within the European Statistical System and the wider international community will be vital to support the responsible and sustainable use of AI in Official Statistics.

References

1. United Nations Economic Commission for Europe: Machine learning for official statistics. Technical report, UNECE, Geneva (2022). URL https://unece.org/sites/default/files/2022-09/ECECESSTAT20216.pdf
2. European Commission: European statistics code of practice. Tech. rep., Eurostat, Luxembourg (2017). URL https://ec.europa.eu/eurostat/web/products-manuals-and-guidelines/-/KS-02-18-142
3. United Nations: Fundamental principles of official statistics. Tech. rep., United Nations General Assembly (2014). URL https://unstats.un.org/unsd/dnss/gp/fundprinciples.aspx. Resolution 68/261
4. UNECE High-Level Group for the Modernisation of Official Statistics: Large language models for official statistics. White paper, UNECE, Geneva (2023). URL https://unece.org/sites/default/files/2023-12/HLGMOS_LLM_Paper_Preprint_1.pdf
5. Massacci, G., Bruno, M., Boselli, C.: Official statistics in the era of ai: Istat's journey towards modern user-centred dissemination systems. Presentation at the UNECE HLG-MOS Workshop on the Modernisation of Official Statistics (2025). URL https://unece.org/sites/default/files/2025-11/HLG2025_Day3_AI_Italy_Massacci_D.pdf
6. Bruno, M., et al.: Big Data and Machine Learning at Istat, pp. 239–298. Springer Nature Switzerland, Cham (2025). DOI 10.1007/978-3-032-10004-7_12. URL https://doi.org/10.1007/978-3-032-10004-7_12
7. D'Amore, G., Summa, D., Amarone, M., et al.: From web to code: Ai-driven product classification for smarter business surveys. Proceedings of the ASA Conference (2025)
8. Tuoto, T., Faricelli, C., Piovani, A.: Tourists' presence flash estimation by mobile network operators data. Rivista Italiana di Economia, Demografia e Statistica **LXXX**(4) (2026). DOI 10.71014/sieds.v80i4.579
9. Bernasconi, E., De Fausti, F., Pugliese, F., Scannapieco, M., Zardetto, D.: Automatic extraction of land cover statistics from satellite imagery by deep learning. Statistical Journal of the IAOS **38**(1), 183–199 (2022). DOI 10.3233/SJI-190555
10. European Statistical System (ESS): Essnet on artificial intelligence and machine learning for official statistics (aiml4os): One-stop shop for ai/ml in official statistics. Eurostat (2024). URL https://cros.ec.europa.eu/dashboard/aiml4os
11. European Statistical System (ESS): Essnet on mobile network operator data – mno-minds: Methodological and infrastructure developments for the use of mobile data in official statistics. Eurostat (2023). URL https://cros.ec.europa.eu/mno-minds
12. European Statistical System (ESS): Essnet web intelligence network (win): Integration of web data for official statistics. Eurostat (2022). URL https://cros.ec.europa.eu/landing-page/web-intelligence-network